FIELD NOTES / 2026

BIBEK GHIMIRE

CYBERSECURITY PROJECT PORTFOLIO

Digital forensics. Network defense. Secure connectivity. Linux systems. Bash automation.

WORK FIRST. CLAIMS SECOND.

Six selected academic and personal lab tracks. Each connects configuration choices to observed validation and documents what remains unproven.

A.A.S. CYBERSECURITY / LAGUARDIA CUNY / GPA 3.53

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GITHUB.COM/BEEBEK-G · LINKEDIN.COM/IN/BIBEKGHIMIRE33

Practical evidence.

Projects highlighted for SOC, digital forensics, network security, and systems roles. Lab results are distinguished from production experience.

01 — DIGITAL FORENSICS

10 investigation reports

Autopsy / communications / recovery / IR

02 — PFSENSE + HYPER-V

3 virtual interfaces configured

LAN, WAN, OPT1 / Ubuntu reachability

03 — TAILSCALE VPN

Subnet advertisement approved

2 connected peers / 192.168.1.0/24

04 — LINUX FIREWALL

Active services + port probes

iptables / SSH / Apache / Netcat

05 — BASH AUTOMATION

6 scripts / CRUD-style operations

Duplicate check / search / delete / list

06 — FEDORA FTP

3 observed successful uploads

vsftpd / firewalld / SELinux

Evidence standard: configurations and test outputs shown in supplied images; no unsupported claims of enterprise deployment, professional incident response, or application reachability.

Investigate. Preserve. Explain.

Ten documented labs from MAC 257. The existing online case links individual report titles and describes each exercise.

10 REPORT-BASED INVESTIGATIONS WINDOWS / LINUX / MOBILE / MEMORY

INVESTIGATION AREAS

01	Evidence admissibility / Daubert	06	Linux forensic artifacts
02	Steganography / hidden data	07	Email and chat logs
03	Deleted and damaged file recovery	08	Mobile device evidence
04	Incident-response investigation	09	Network infrastructure evidence
05	Windows forensic artifacts	10	System memory examination

METHOD OVER TOOLS

Describe the question, preserve a clean evidence trail, record tools and methodology, report relevant findings and distinguish observation from inference. Academic lab outcomes are not presented as legal case conclusions.

CASE STUDY: GITHUB.COM/BEEBEK-G/DIGITAL-FORENSICS-LABS

A virtual perimeter.

pfSense CE 2.8.1 on Hyper-V with distinct interfaces and validation from an Ubuntu client.

WAN

192.168.1.178

DHCP / hn0

LAN

192.168.0.1

/24 / hn1

OPT1

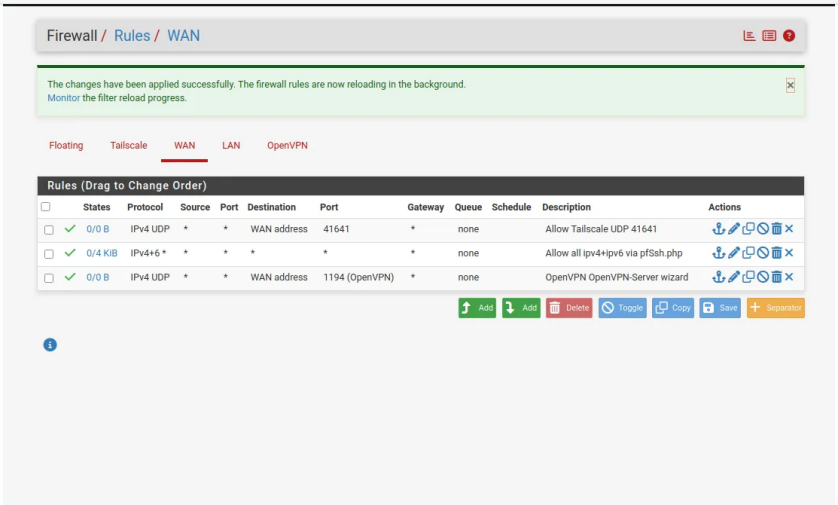
10.168.153.1

/24 / hn2

WHAT THE TESTS SHOWED

- Ubuntu client reached its LAN gateway: 4 sent / 4 received, 0% loss.
- Public test address responded: 4 / 4, 0% loss; OPT1 test host: 5 / 5, 0% loss.
- Rule edits and an applied-change message were visible in the pfSense UI.

EXAMPLE CONFIGURATION SCREEN



Boundary: reachability tests do not prove full segmentation. Some screenshots show broad allow rules; production policy should be limited to specific required flows.

Beyond the local subnet.

Tailscale on pfSense: peer status, firewall access, route advertisement and admin approval.

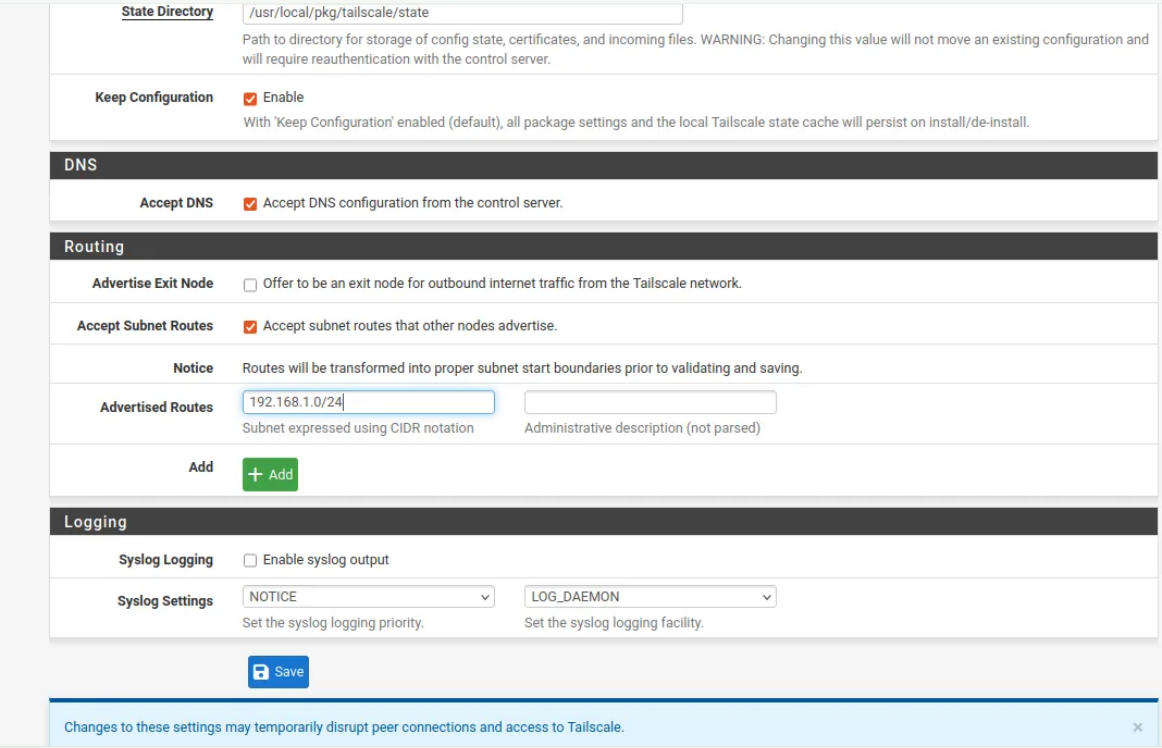
TWO CONNECTED NODES

PFSENSE / FREEBSD

IOS / MOBILE PEER

ADVERTISED + APPROVED SUBNET 192.168.1.0/24

CONFIGURATION EVIDENCE



Interpretation: status pages verify joined peers and an approved subnet route. The provided screenshots do not demonstrate a mobile-to-LAN application request. Sensitive authentication material is deliberately excluded from public artifacts.

Test the boundary.

Ubuntu iptables and service state, then Fedora vsftpd provisioning and file-transfer results.

UBUNTU / IPTABLES

SSH + APACHE

Started SSH and Apache2 and confirmed active state. Tested INPUT policy changes with Netcat to TCP/80 and compared ACCEPT versus REJECT behavior.

VALIDATION

Apache default page loaded locally. Netcat successful / refused outcomes were compared after rule edits; kernel logs showed a custom prefix.

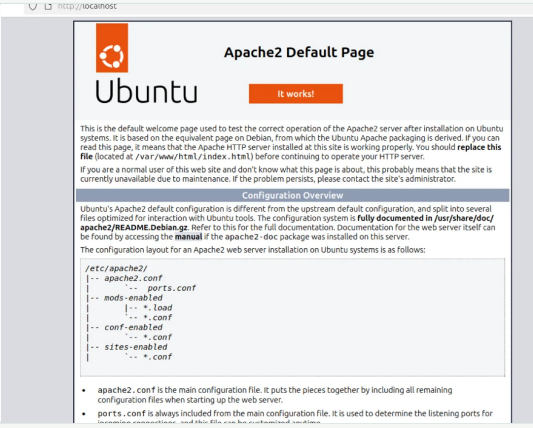
FEDORA / VSFTPD

THREE UPLOADS

Configured local authentication, disabled anonymous FTP, enabled writes and checked firewalld / SELinux exercise settings.

VALIDATION

One `put` plus two `mput` operations each returned transfer complete. A later `get` failed with a 550 response.



Operational boundary: exercises used virtual machines and localhost. Plain FTP and overly broad firewall/SELinux access are not recommended for sensitive production workloads.

Make it inspectable.

A command-line jobs database built with Bash, AWK and a plain-text record store.

MAC232 FINAL PROJECT / SHELL PROGRAMMING

01	ADD	case-insensitive duplicate JobID check
02	SEARCH	find by field four with AWK
03	REPORT	formatted multi-column list
04	DELETE	rewrite excluding matching JobID
05	RAW FILE	print current colon-delimited records

WHAT THE SCREENSHOTS CONFIRM

- Existing JobID rejected; new JobID added, found by search, displayed and then deleted.
- Project tree included six executable scripts, a data file and pseudocode documentation.
- Hardening backlog: field validation, safe temporary files, atomic replacement and test coverage.

ABOUT

A.A.S. Cybersecurity, LaGuardia Community College, CUNY (2026; GPA 3.53). Currently enrolled at John Jay College of Criminal Justice, CUNY, studying Computer Science and Information Security. Prior work: web design and IT support at DIT Solution, Kathmandu.

CONNECT / REQUEST MORE EVIDENCE

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